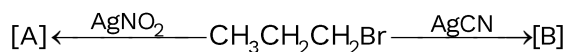


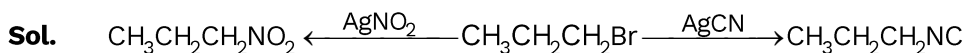
CHEMISTRY

Q1 The product A & B in the following reaction respectively:



- (1) $\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—ONO}$, $\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—CN}$
 (2) $\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—NO}_2$, $\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—NC}$
 (3) $\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—NO}_2$, $\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—CN}$
 (4) $\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—ONO}$, $\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—NC}$

Ans. (2)

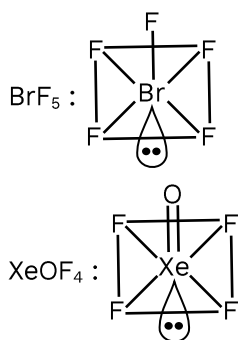


Q2 The molecules having square pyramidal geometry are

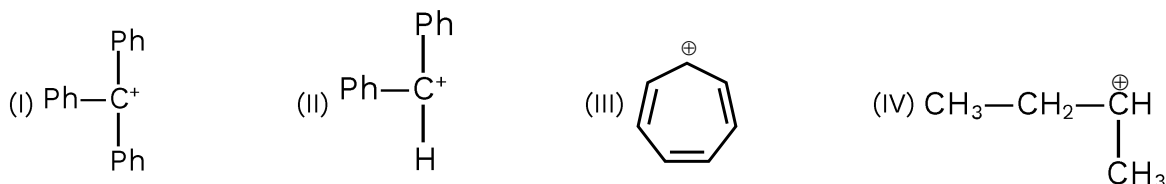
- (1) SbF_5 & PCl_5 (2) BrF_5 & XeOF_4 (3) BrF_5 & PCl_5 (4) SbF_5 & XeF_4

Ans. (2)

Sol.



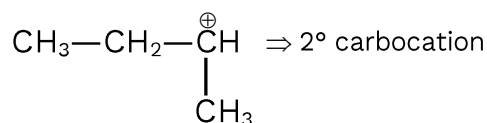
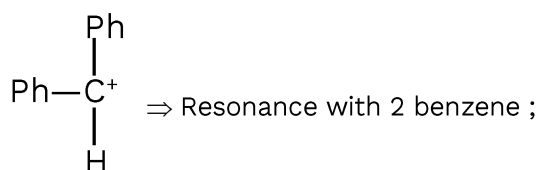
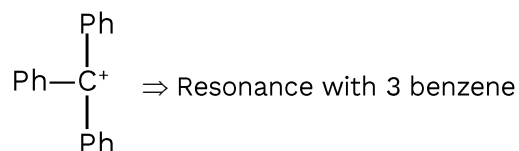
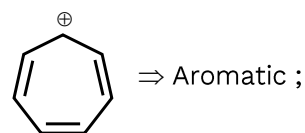
Q3 Increasing order of stability of following carbocation is:



- (1) I > II > III > IV (2) III > I > II > IV (3) II > I > IV > III (4) II > III > I > IV

Ans. (2)

Sol.



Q4 From the following element Al, Ga, Tl and Pb. The most stable oxidation states of elements with highest and lowest first ionization and enthalpies respectively are

- (1) +2 and +3 (2) +4 and +3 (3) +1 and +4 (4) +4 and +1

Ans. (1)

Sol. Ionisation energy : Pb > Tl > Ga > Al

Highest Ionisation energy : Pb²⁺

Q5 The incorrect decreasing order of atomic radii is

- (1) Be > Mg > Al > Si (2) Al > B > N > F (3) Si > P > Cl > F (4) Mg > Al > C > O

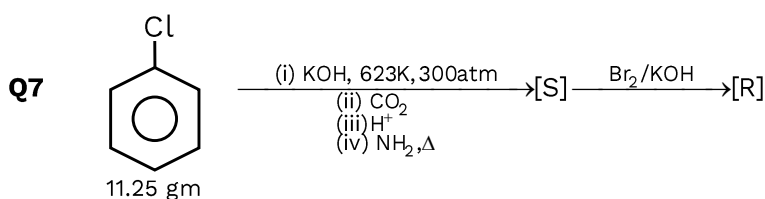
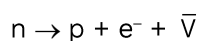
Ans. (1)

Q6 Which of the following given reaction is correct

- (1) $n \rightarrow p + e^- + \bar{\nu}$ (2) $n \rightarrow p + e^+ + \bar{\nu}$ (3) $n \rightarrow p + e^+ + \nu$ (4) $n \rightarrow p + e^- + \bar{\nu}$

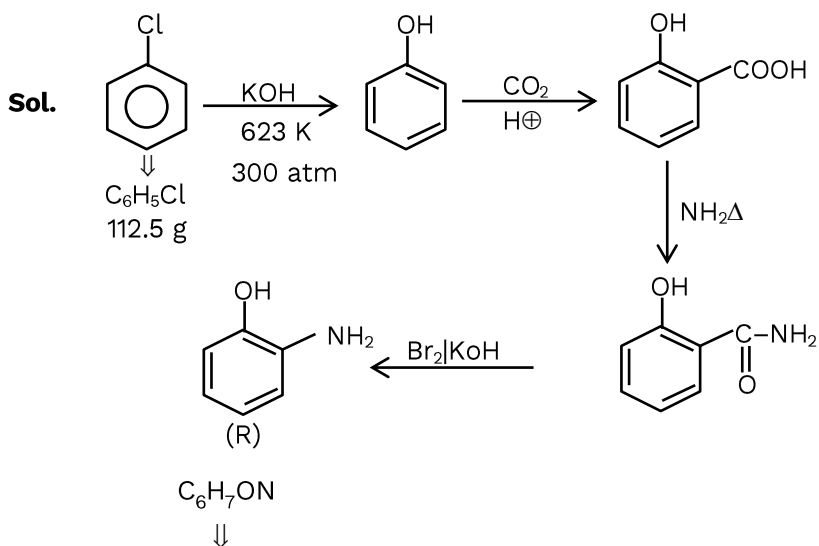
Ans. (4)

Sol. Unstable nucleus decrease the ratio n—p to obtain the stability.



Find the mass of the product [R]:

Ans. (10.9)



$$6 \times 12 + 7 \times 1 + 16 + 14$$

$$72 + 7 + 16 + 14 = 109 \text{ g}$$

112.5 g Chlorobenzene $\rightarrow$ 109 g [R]

$$11.25 \text{ g will give } \rightarrow \frac{109}{112.5} \times 11.25 = 10.9 \text{ gm}$$

Q8 Which of the following set of quantum number have same energy?

- (a) $n = 2, l = 0, m = +1$ (b) $n = 2, l = 1, m = -1$
 (c) $n = 3, l = 2, m = 0$ (d) $n = 3, l = 2, m = 1$
 (1) a, b (2) c, d (3) b, c (4) a, c

Ans. (2)

Sol. (c) $n = 3, l = 2, m = 0 \Rightarrow 3d$

(d) $n = 3, l = 2, m = 1 \Rightarrow 3d$

Due to same n & l they are degenerate (same energy)

Q9 Given below are two statements:

Statement (I): Pentacetate/Glucose react with 2,4-DNP.

Statement (II): Starch react with sulphuric acid at given temperature with 2 atm pressure gives glucose.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement I and Statement II are incorrect
 (2) Statement I is correct but Statement II is incorrect
 (3) Both Statement I and Statement II are correct
 (4) Statement I is incorrect but Statement II is correct

Ans. (4)

Sol. Conceptual

Q10 CH_3CHO and $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$

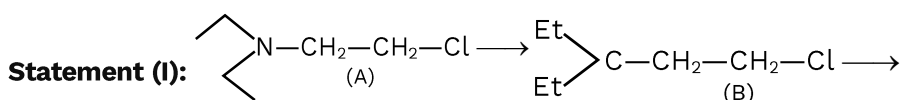
How many reagents give both the test?

- (a) Tollen's Test (b) Iodoform Test (c) Aldol
 (d) Cannizaro (e) Fehling's Test

Ans. (2)

Sol. b & c

Q11 Given below are two statements:



(A) give hydrolysis faster than (B).

Statement (II): Compound (A) first combined itself to give intramolecular bond.

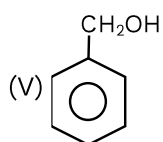
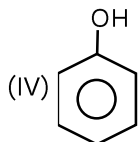
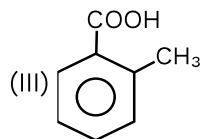
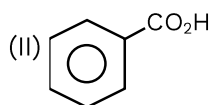
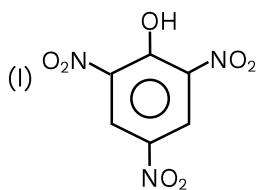
In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement I and Statement II are incorrect
 (2) Statement I is correct but Statement II is incorrect
 (3) Both Statement I and Statement II are correct
 (4) Statement I is incorrect but Statement II is correct

Ans. (4)

Sol. Conceptual

Q12 What is the rate of reaction for releasing $\text{CO}_2(\text{g})$ with aq. NaHCO_3 among the following?



(1) $\text{I} > \text{II} > \text{III} > \text{IV} > \text{V}$ (2) $\text{I} > \text{III} > \text{IV} > \text{II} > \text{V}$ (3) $\text{I} > \text{III} > \text{II} > \text{IV} > \text{V}$ (4) $\text{I} > \text{II} > \text{III} > \text{V} > \text{IV}$

Ans. (3)

Sol. Conceptual

Q13 Which of the following ions gives violet colour in borax bead test in hot flame and has no effect on type of complex

(1) Ni^{2+}

(2) Co^{2+}

(3) Mn^{2+}

(4) Fe^{2+}

Ans. (1,3)

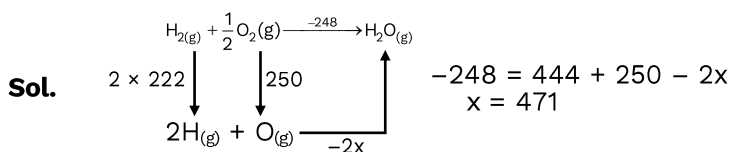
Sol. $\text{Ni}^{2+} : 3d^8 \Rightarrow sp^3d^2$ (Octahedral) $\rightarrow$ Violet Colour

$\text{Mn}^{2+} : 3d^5 \Rightarrow$ Octahedral $\begin{cases} \rightarrow \text{Under SFL} : d^3sp^3 : 1 \text{ unpaired } e^- \\ \rightarrow \text{Under WFL} : sp^3d^2 : 5 \text{ unpaired } e^- \end{cases} \rightarrow \text{Light Violet Colour}$

Q14 $\Delta_f H$ of $\text{H}(\text{g})$ is 222 kJ/mol,
 $\Delta_f H$ of $\text{O}(\text{g})$ is 250 kJ/mol,
 $\Delta_f H$ of H_2O is -248 kJ/mol.

What is the value of bond energy of O-H bond in H_2O in kJ/mol?

Ans. (471)



Q15 70% by mass solution of HNO_3 is taken, having density 1.41 gm/ml Calculate molarity.
 (Rounded off to nearest integer)

Ans. (16)

Sol. Solution 100 gm = $\frac{(100)}{(1.41)}$ ml

$\text{HNO}_3 = 70 \text{ gm} = \left(\frac{70}{63} \right) \text{ mole}$

$$\text{Molarity} = \frac{\left(\frac{70}{63}\right)}{\frac{0.1}{1.41}} = 15.66\text{M} \approx 16\text{ M}$$

Q16 Match the List:

List-I

- (P) Decomposition reaction
 (Q) Combustion reaction
 (R) Displacement reaction
 (S) Disproportionation reaction
 (1) $P \rightarrow C$; $Q \rightarrow B$; $R \rightarrow D$; $S \rightarrow A$
 (3) $P \rightarrow C$; $Q \rightarrow D$; $R \rightarrow B$; $S \rightarrow A$

List-II

- (A) $\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$
 (B) $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
 (C) $\text{NaH} \rightarrow \text{Na} + \text{H}_2$
 (D) $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$
 (2) $P \rightarrow B$; $Q \rightarrow C$; $R \rightarrow D$; $S \rightarrow A$
 (4) $P \rightarrow C$; $Q \rightarrow B$; $R \rightarrow A$; $S \rightarrow D$

Ans. (1)

Sol. Factual based

Q17 $\text{K}_2\text{Cr}_2\text{O}_7 + \text{KMnO}_4$

On reaction with $\text{K}_2\text{Cr}_2\text{O}_7$ or KMnO_4 in acidic medium. Which of the following reaction is possible.

- (A) $\text{I}^- \rightarrow \text{I}_2$ (B) $\text{S}^{2-} \rightarrow \text{S} \downarrow$ (C) $\text{S}_2\text{O}_3^{2-} \rightarrow \text{SO}_4^{2-}$
 (D) $\text{I}^- \rightarrow \text{IO}_3^-$ (E) $\text{Fe}^{2+} \longrightarrow \text{Fe}^{3+}$
 (1) A, B & E Only (2) B & C Only (3) A, B & C Only (4) C & D Only

Ans. (1)

Sol. NCERT based

Q18 Number of lone pairs in equatorial position of ClF_3 is equal to number of unpaired electron of Ni^{+2} , Mn^{+2} , Cu^{+2} , Ti^{+2} , V^{+3}

- (1) Ni^{+2} , Mn^{+2} , Cu^{+2} (2) Mn^{+2} , Cu^{+2} (3) Ni^{+2} , Cu^{+2} , V^{+3} (4) Ni^{+2} , Mn^{+2} , V^{+3}

Ans. (4)

Sol. Ni^{2+} , V^{3+} , Ti^{2+} Correct

- $\text{Ni}^{+2} = 3d^8 \rightarrow 2 \text{ unpaired } e^-$
 $\text{Mn}^{+2} = 3d^5 \rightarrow 5 \text{ unpaired } e^-$
 $\text{Cu}^{+2} = 3d^9 \rightarrow 1 \text{ unpaired } e^-$
 $\text{Ti}^{+2} = 3d^2 \rightarrow 2 \text{ unpaired } e^-$
 $\text{V}^{+3} = 3d^2 \rightarrow 2 \text{ unpaired } e^-$

